

$^4\text{He}(^{30}\text{S},\alpha)$:resonances **2018Ka11**

$^{30}\text{S}+\alpha$ resonant elastic scattering. $J^\pi=0^+$ for ^{30}S g.s. and ^4He g.s.

2018Ka11: 48.4-MeV ^{20}S was produced via the $^3\text{He}(^{28}\text{Si},^{30}\text{S})\text{n}$ reaction, separated by the CRIB separator at RIKEN, passed through two PPACs, and entered an active target filled with 90% $^{\text{nat}}\text{He}$ + 10% CO_2 gas at 194.2 Torr 5. Scattered α particles were detected using the active target and 500- μm thick SSDs. Measured E_α , I_α , and $^{30}\text{S}+\alpha$ excitation functions at $E_{\text{c.m.}}=2.0\text{--}5.5$ MeV. Deduced 3 resonances, J , π , L_α , Γ_α , and Γ_p from multichannel R-Matrix analysis.

 ^{34}Ar Levels

<u>$E(\text{level})^\dagger$</u>	<u>$J^\pi^\dagger$</u>	<u>$L^\dagger$</u>	<u>Comments</u>
11092 85	($2^+, 4^+$)	(2,4)	$E_{\text{c.m.}}=4353$ keV. For $J^\pi=2^+$, $\Gamma_\alpha=20$ keV +80-17, $\Gamma_p=25$ keV +500-20. $\omega\gamma(\alpha,p)=100$ keV. For $J^\pi=4^+$, $\Gamma_\alpha=0.5$ keV +14-4, $\Gamma_p=0.3$ keV +35-3.
11518 89	(2^+)	(2)	$E_{\text{c.m.}}=4779$ keV. $\Gamma_\alpha=100$ keV +120-60, $\Gamma_p=210$ keV +600-170. $\omega\gamma(\alpha,p)=500$ keV.
12079 95	(2^+)	(2)	$E_{\text{c.m.}}=5340$ keV. $\Gamma_\alpha=260$ keV +400-120, $\Gamma_p=340$ keV +550-200. $\omega\gamma(\alpha,p)=1300$ keV.

† From R-Matrix analysis of the measured $^{30}\text{S}+\alpha$ elastic scattering excitation function in **2018Ka11**. The parentheses on L_α are added by evaluators. The parentheses on J^π of the 11518 level are added by evaluators. The selection rules dictate a unique resonance J^π for each L_α value, namely that $J = L_\alpha$ and the parity is always natural for populated states in ^{34}Ar .